Innovative Solutions and Support Inc (IA)

The Avionics Insourcing Inflection Creates an Asymmetric Entry Point

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Innovative Solutions and Support Inc stands at the threshold of a multi-year cash flow compounding cycle as the completion of its Exton facility expansion and strategic manufacturing insourcing permanently elevate structural profitability. The broader market remains overly preoccupied with quarterly defense shipment lumpiness and acquisition integration costs, failing to recognize that commercial aviation momentum and high-margin aftermarket installations are structurally recalibrating baseline operating margins. As operational leverage takes hold and free cash flow conversion accelerates from historical troughs toward normalized levels, the company offers exceptional fundamental upside. We initiate coverage with a Buy rating and a probability-weighted blended price target of $45.67, representing +133.5% upside from the reference spot price of $19.56.

BUY	$45.67	+133.5%	$19.56
INVESTMENT RATING	BLENDED TARGET	BLENDED UPSIDE	REFERENCE PRICE

While consensus remains distracted by temporary quarterly defense revenue lumpiness and integration-related operating expense increases, our quantitative analysis reveals an avionics franchise achieving structurally higher gross margins above 50% following the insourcing inflection at Exton. The market prices Innovative Solutions and Support Inc as a volatile micro-cap defense subcontractor rather than a premier cockpit modernization platform with expanding commercial run-rates and expanding free cash flow conversion.

EXECUTIVE SUMMARY

Innovative Solutions and Support Inc operates as an established avionics enterprise specializing in the design, engineering, and manufacturing of flight-critical systems. The company develops advanced cockpit displays, flight management systems, and engine monitoring products for aircraft across commercial transport, business aviation, and defense platforms. Its operational model has undergone a foundational evolution, shifting from reliance on third-party assembly toward internal manufacturing at its expanded Exton facility. This strategic posture enables the firm to capture enhanced margins, preserve intellectual property, and secure sole-source positions across high-utilization retrofit and line-fit flight decks globally.

The recent quarterly evidentiary record confirms the efficacy of this manufacturing pivot while highlighting strong commercial execution. In the third quarter of FY2026, net sales expanded 10.7% year-over-year as strength in

commercial and business aviation successfully overcame demanding prior-year comparisons associated with previous defense pull-forwards. Gross margin held firmly at 51.7%, markedly outperforming management guidance in the mid-40% range and driving adjusted EBITDA growth of 74.7%. This followed the second quarter of FY2026, where non-defense commercial revenue surged 69% year-over-year, insulating the consolidated enterprise from delayed IPDG regulatory approvals on defense programs and demonstrating resilient baseline demand for flagship cockpit avionics.

Capital allocation priorities have balanced continuous operational reinvestment with disciplined inorganic expansion. During the recent fiscal period, management concluded the primary capital expenditure cycle at Exton, directly catalyzing a 155% year-over-year surge in nine-month free cash flow generation. The firm utilized its enhanced operating cash flow, which grew 50.4% over the first nine months of FY2026, alongside conservative debt financing to complete three targeted acquisitions, including the closing of the Aydin Displays transaction in July 2026. While inorganic integration and specialized research and development elevated third-quarter operating expenses by 53% year-over-year, net leverage remains modest at 1.4x, leaving ample liquidity to support ongoing platform growth.

For the full fiscal year FY2025, Innovative Solutions and Support Inc generated revenue of $84.3M, representing an exceptional 78.6% year-over-year increase, with gross profit of $40.5M and EBIT of $20.1M. Operating margin reached 23.8% while net income totaled $15.6M, yielding diluted earnings per share of $0.88 and a return on equity of 24.2%. Free cash flow concluded the fiscal year at $6.8M, reflecting working capital commitments and safety-stock inventory builds ahead of the insourced manufacturing ramp. With total debt of $24.1M, cash of $2.7M, and net debt of $21.4M at the close of FY2025, the company maintains a solid balance sheet underpinning our high-conviction positive outlook.

OPERATIONAL SNAPSHOT

$19.56	$84.3M	$6.8M	8.1%	0.60	$19.56
REF. PRICE	FY2025 REVENUE	FY2025 FREE CASH FLOW	FCF MARGIN	EQUITY BETA	REF. CLOSE

Fiscal-year figures as reported in company filings and collected by Cascadia Analytics; ref. price as of 2026-10-20.

THE MOAT IS SPECIALIZED AND ENTRENCHED

Innovative Solutions and Support Inc maintains a wide competitive moat anchored in high customer switching costs and stringent regulatory barriers to entry. The avionics industry requires exhaustive certification through the Federal Aviation Administration and international civil aviation authorities. Cockpit displays, flight management systems, and engine monitoring products undergo rigorous qualification processes that frequently require multi-year validation cycles. Once an avionics suite is certified and integrated into an aircraft type certificate or supplemental type certificate, airframe operators face prohibitive engineering expenses, downtime costs, and pilot re-certification hurdles if they attempt to displace an incumbent supplier. This regulatory friction creates durable customer lock-in spanning decades-long aircraft operational lifecycles.

The company reinforces this competitive position through sole-source proprietary intellectual property. By designing custom flat-panel cockpit displays and specialized flight management software algorithms, Innovative Solutions and Support Inc embeds its hardware and software directly into the primary pilot interface. Its engine monitoring units deliver high-precision sensor processing that is critical for operational safety and fuel optimization. In business aviation and commercial transport retrofit programs, operators select these architectures because they allow older airframes to comply with modernized airspace mandates without requiring wholesale structural rewiring. The engineering know-how required to interface modern digital processing with legacy analog aircraft sensors constitutes a significant barrier that deter generic competitors.

Operational insourcing at the Exton facility has expanded the durability of this franchise moat. Historically, contract manufacturing introduced margin leakage and external supply chain vulnerabilities. By bringing manufacturing, assembly, and testing in-house, the firm has achieved vertical integration that protects proprietary manufacturing techniques, compresses production lead times, and expands structural gross margins. This operational capability was clearly demonstrated when gross margins expanded 1,310 basis points year-over-year to 54.5% in the first quarter of FY2026, driven directly by insourcing efficiencies. Competitors relying on external merchant suppliers cannot readily match this combination of cost advantage and bespoke engineering flexibility.

Customer retention is further buttressed by an expanding installed base that yields recurring, high-margin aftermarket component replacement and maintenance demand. Aircraft operating in commercial transport, business fleets, and defense squadrons require continuous spares provisioning and periodic software updates. Because safety certifications mandate that replacement parts match original manufacturer specifications, Innovative Solutions and Support Inc captures steady aftermarket demand with minimal incremental sales friction. This structural dynamic provides strong pricing power, as demonstrated by the company sustaining gross margins above 51% throughout recent quarters despite volatile macroeconomic supply chain pressures.

VALUATION

Three scenarios make the valuation range explicit

Our valuation framework uses a multi-stage discounted cash flow model calibrated to explicit fundamental drivers across a ten-year forecast window from FY2026 through FY2035. We model the trajectory of the firm as it transitions from its recent capital investment cycle into a normalized cash generation profile. Baseline revenue compounding is driven by steady adoption of cockpit displays, expanding flight management system installations, and inorganic contributions from integrated acquisitions. The terminal valuation applies an explicit terminal growth rate of 3.0% under our base case, discounted at a weighted average cost of capital of 9.33% across 17.9 million diluted shares.

The critical driver of shareholder value creation in our model is the normalization of free cash flow conversion. During FY2025, free cash flow was $6.8M on revenue of $84.3M, reflecting a free cash flow margin of 8.1% that was depressed by inventory builds and the capital-intensive Exton facility buildout. As capital expenditures abate and operating leverage scales against corporate overhead, free cash flow margins normalize toward long-term targets across all scenarios. The market capitalization of $350.1M at the reference spot price of $19.56 discounts an overly pessimistic outcome, creating a highly favorable risk-reward asymmetry.

SCENARIO	WEIGHT	REVENUE CAGR	TARGET FCF MARGIN	TARGET	UPSIDE
Bear	25%	11.7%	15.2%	$19.20	-1.8%
Base	50%	18.4%	18.0%	$45.36	+131.9%
Bull	25%	21.5%	20.0%	$72.77	+272.0%

Bear $19.20

The bear scenario assumes an 11.7% revenue compound annual growth rate through FY2035 with the target free cash flow margin reaching 15.2%. This path reflects persistent defense approval delays, protracted integration friction from acquired entities, and higher sustained operating expenses that restrict operating leverage. The resulting equity target is $19.20, representing limited downside of -1.8% from current trading levels.

Base $45.36

Our base case projects an 18.4% revenue compound annual growth rate and a normalized target free cash flow margin of 18.0% at the end of the forecast window. Steady commercial aviation retrofit demand, resolution of regulatory hurdles, and structural manufacturing efficiencies at Exton unlock substantial operational leverage. This scenario generates an intrinsic equity value of $45.36 per share, representing +131.9% upside.

Bull $72.77

The bull case models an accelerated 21.5% revenue compound annual growth rate and an optimized 20.0% target free cash flow margin. Rapid integration of acquired product lines, accelerated market share gains in commercial transport cockpit displays, and sustained gross margins above 55% drive exceptional cash compounding. This optimistic scenario yields a target of $72.77, offering +272.0% upside.

VALUATION FRAMEWORK

A live discount rate, not a static hurdle

We evaluate Innovative Solutions and Support Inc using a discount rate of 9.33%, derived through a standard capital asset pricing model framework that reflects the company's financial and operating profile. The model incorporates an equity beta input of 0.60, capturing the defensive characteristics of long-cycle aerospace programs, substantial regulatory barriers, and sole-source aftermarket positions. While small-capitalization equities typically carry elevated market volatility, the company's contracted backlog, absence of speculative inventory risk, and low net debt mitigate fundamental operating exposure.

The 9.33% discount rate accurately balances macroeconomic cost of capital pressures against the enterprise's high return on equity, which reached 24.2% in FY2025. It reflects a realistic hurdle rate for an avionics enterprise executing an inorganic expansion strategy while maintaining net leverage of 1.4x. Rather than applying an arbitrary hurdle rate, this dynamic cost of capital explicitly captures

KEY VALUATION PARAMETERS

Forecast window	**FY2026 to FY2035**
Live WACC	**9.33% (2026-10-20)**
Terminal growth	**3.0%**
Equity beta	**0.60**
Diluted shares	**17.9 million**

BUY

market risk while appropriately rewarding the business for completing its capital expenditure cycle at Exton.

FCF margin normalization is the engine

The central engine of our equity thesis is the substantial expansion of free cash flow margins from historical levels toward our terminal targets. In FY2025, the company achieved an operating margin of 23.8% and EBITDA of $25.4M, yet free cash flow margin was constrained at 8.1%, yielding $6.8M in free cash flow. This divergence was driven by working capital investments, safety-stock inventory accumulation, and peak facility outlays associated with insourcing production. With the Exton expansion completed, capital requirements have decreased substantially, enabling cash conversion to align more closely with operating profitability.

This operational transition is already visible in the evidentiary record, as nine-month free cash flow expanded 155% year-over-year in FY2026. Over our forecast window, free cash flow margins normalize from 8.1% to 15.2% in the bear case, 18.0% in the base case, and 20.0% in the bull case. These targets are fully supported by the company's demonstrated gross margin durability, which reached 51.7% in Q3 FY2026 and 54.5% in Q1 FY2026, alongside corporate operating expense discipline as integration activities conclude.

Mechanics are explicit

Revenue becomes EBIT, cash taxes come out, then we add back D&A, subtract CapEx and the change in net working capital to reach unlevered free cash flow, which we discount and divide by diluted shares.

Methodology. Detailed EBIT-bridge DCF on three scenarios, probability-weighted 25/50/25. Beta decays toward a mature peer over the forecast. Terminal value via Gordon growth. Targets are model-computed, not estimated.

over $45.67

blended target

BUSINESS & SEGMENT DRIVERS

Revenue prospects anchor on Cockpit Modernization and Commercial Aftermarket Expansion BUY

Innovative Solutions and Support Inc organizes its commercial activities across integrated avionics categories, specifically cockpit displays, flight management systems, and engine monitoring products. Cockpit displays represent the core visual and processing hub for flight crews, providing critical primary flight telemetry, navigation mapping, and engine instrument crew alerting systems. Demand for these displays is propelled by fleet

modernization programs across commercial air transport and business aviation, where operators seek weight reduction, enhanced situational awareness, and compliance with next-generation airspace navigational criteria.

The flight management systems portfolio serves as the intellectual computational core of aircraft navigation. These systems automate flight planning, route optimization, performance monitoring, and lateral and vertical guidance. Growth across this line is reinforced by rising demand for reduced fuel burn and optimized arrival trajectories, which require advanced algorithmic processing. Because flight management software is deeply integrated with cockpit displays and auto-throttle architectures, the company frequently captures multi-product bundle wins that enhance revenue per airframe while locking in long-term maintenance cycles.

Engine monitoring products constitute the third primary pillar of the operating model, providing continuous acquisition and diagnostic processing of critical propulsion parameters. These units interface high-temperature, high-vibration sensor inputs with digital flight deck management architectures. In business aviation and specialized transport aircraft, accurate engine health tracking reduces catastrophic engine failures and optimizes scheduled overhaul intervals. As aircraft operators prioritize predictive maintenance to maximize fleet uptime, engine monitoring products provide dependable, high-margin shipments that complement cyclic flight deck retrofit programs.

The operational trajectory documented across recent quarters highlights a structural rebalancing between commercial aviation and defense applications. While defense shipments experienced volatility due to delayed regulatory approvals, the commercial business demonstrated robust underlying momentum, surging 69% year-over-year in the second quarter of FY2026. Furthermore, strategic acquisitions, including the purchase of Aydin Displays completed in July 2026, enhance technical capabilities and expand customer channels across military and specialized display markets. Supported by high gross margins and an ending backlog position that supports forward production visibility, the enterprise remains exceptionally well-positioned to execute its long-term growth roadmap.

What to monitor

- **Commercial aviation delivery cadence.** Monitor whether commercial and business aviation product lines maintain robust shipment growth to continuously offset defense revenue lumpy periods.
- **Gross margin stability above fifty percent.** Track post-insourcing gross margins relative to historical mid-40% guidance to confirm that Exton manufacturing efficiencies remain intact.
- **Operating expense absorption and integration.** Observe quarterly selling, general, and administrative expenses alongside research and development outlays to verify that acquisition integration expenses normalize.
- **Regulatory milestone clearance.** Assess the timing of delayed defense approvals to determine when deferred government program shipments will re-accelerate consolidated volume.

+78.6%	48.1%	23.8%	8.1%	$21.4M	24.2%
FY2025 REVENUE GROWTH	GROSS MARGIN	OPERATING MARGIN	FCF MARGIN	NET DEBT	RETURN ON EQUITY

METRIC	FY2025 (AS REPORTED)
Revenue	$84.3M

Gross Profit	$40.5M
EBIT	$20.1M
Net Income	$15.6M
EBITDA	$25.4M
Free Cash Flow	$6.8M
Diluted EPS	$0.88
ROE	24.2%

RISK & MOAT MATRIX

The bear case

SHARE EROSION

Tier-one avionics conglomerates could intensify competitive pricing across commercial retrofit programs to pressure smaller independent suppliers. Innovative Solutions and Support Inc counters this threat through bespoke engineering integration and lower total retrofit downtime for aircraft operators.

M&A DIGESTION

Rapid execution of acquisitions, including the recent closing of Aydin Displays, could result in prolonged operational friction and elevated overhead costs. Extended integration timelines would weigh on consolidated operating margins if expected synergies fail to

REGULATORY AND PRICING SCRUTINY

Delays in obtaining certification approvals from civil aviation authorities or defense procurement agencies can defer high-margin product deliveries into future periods. Regulatory friction directly impacts quarterly revenue predictability, as demonstrated by recent timing shifts in defense shipment schedules.

TECHNOLOGY DISRUPTION

Accelerating migration toward integrated modular avionics architecture requires continuous research and development commitments to avoid product obsolescence. Failure to maintain software compatibility and cutting-edge display performance could erode the firm's technological positioning over time.

REINVESTMENT TREADMILL

Sustaining state-of-the-art internal manufacturing capabilities at Exton may require ongoing capital allocation for precision testing and assembly automation. If maintenance capital expenditures exceed projected run-rates, long-term free cash flow margin conversion would be impaired.

WACC AND DURATION SENSITIVITY

A higher-for-longer macroeconomic interest rate environment would elevate capital costs and compress valuation multiples for growth-oriented aerospace suppliers. Given the long-duration cash flow profile of our discounted cash flow model, elevated discount rates represent a core financial sensitivity.

	materialize promptly.				

The strengths

OLIGOPOLY POSITION

The firm commands an entrenched position in specialized retrofit cockpit displays and flight management architectures protected by rigorous certification barriers. High switching costs ensure that incumbent airframe positions remain secure throughout multi-decade aircraft lifecycles.

FCF COMPOUNDING

With the major capital expenditure cycle at Exton concluded, operating cash flow converts efficiently into free cash flow. This structural inflection supports rapid debt deleveraging and self-funded growth without requiring dilutive equity financing.

GEOGRAPHIC DURABILITY

An extensive international footprint across commercial airline operators, cargo carriers, and business fleets provides diversified end-market demand. Global regulatory airspace modernization mandates ensure steady ongoing retrofits regardless of regional macroeconomic fluctuations.

PIPELINE OPTIONALITY

Vertically integrated internal production lines allow the firm to rapidly engineer custom flight deck solutions for newly acquired product lines. This manufacturing flexibility enhances the accretion of acquired entities like Aydin Displays while expanding total addressable content per airframe.

The bottom line on risk and reward

The balance of fundamental risks and operational catalysts for Innovative Solutions and Support Inc tilts overwhelmingly to the upside. The bear scenario, which incorporates prolonged defense program softness and compressed margins, yields an intrinsic value of $19.20 per share, establishing that the current market price of $19.56 already reflects substantial pessimism. Downside risk appears exceptionally well-buffered by the company's contracted backlog, solid 24.2% return on equity, and low net debt position.

Conversely, our base scenario of $45.36 and bull scenario of $72.77 demonstrate the extraordinary shareholder value unleashed when insourced manufacturing efficiencies take full effect. As the Exton facility continues to drive gross margins above 50% and free cash flow conversion expands from the 8.1% recorded in FY2025 toward normalized levels of 18.0%, institutional rerating will follow. The valuation disconnect created by temporary quarterly defense shipment lumpiness provides a rare asymmetric entry point into an elite, sole-source avionics engineering franchise.

Weighting our scenarios at 25% bear, 50% base, and 25% bull produces a blended price target of $45.67. This target implies an upside potential of +133.5% from the reference spot price of $19.56. With superior operating leverage, expanding commercial aviation market share, and an inflecting free cash flow profile, we reiterate our high-conviction Buy recommendation on the shares of Innovative Solutions and Support Inc.

Justin Wen, Senior Equity Research Analyst, Cascadia Analytics

FRANCHISE DEEP DIVE

Cockpit Displays and Flight Management Systems: Structural Flight Deck Modernization

Innovative Solutions and Support Inc has established an industry-leading reputation for developing high-resolution, sunlight-readable cockpit flat-panel displays and tightly integrated flight management software suites. These systems replace antiquated electromechanical instruments and cathode-ray tube screens with modern digital flight instruments, multi-function displays, and computerized navigation interfaces. By providing high-reliability flight deck modernization packages, the company enables aircraft owners to modernize older aircraft at a fraction of the cost required to acquire new airframes. This value proposition has driven widespread adoption across both commercial transport and business aviation fleets.

The technological advantage of the company's cockpit displays resides in their low power consumption, minimal heat generation, and low overall unit weight, which directly translates into reduced fuel burn for operators. Coupled with proprietary flight management systems, these platforms provide synthetic vision, graphical flight planning, and advanced navigation capabilities. The operational integration between display hardware and software management engines ensures seamless situational awareness during all phases of flight, reducing pilot workload and significantly elevating flight safety standards.

Furthermore, the firm's engine monitoring products complement the flight deck display suite by gathering, filtering, and transmitting real-time propulsion data to central flight displays. This complete situational monitoring loop creates an integrated ecosystem where airframe operators rely on Innovative Solutions and Support Inc for their entire primary flight information interface. The resulting ecosystem generates strong brand loyalty and recurring revenue streams through software updates, system enhancements, and certified aftermarket replacement hardware throughout the operational life of the aircraft.

Recent operational performance confirms that this flagship franchise is experiencing accelerated commercial adoption. The insourcing of manufacturing operations into Exton has transformed unit economics, permitting faster delivery timelines and improved quality assurance while structurally lifting gross margins. As airlines and corporate fleet operators accelerate modernization programs to maintain compliance with evolving airspace requirements, Innovative Solutions and Support Inc possesses the technological depth, certified product portfolio, and manufacturing scale necessary to capture sustained, highly profitable market share.

What the recent record shows

- Gross margin durability held above 51% throughout recent quarters, significantly exceeding management's historical mid-40% guidance, driven by commercial aftermarket mix and the completion of the Exton insourcing transition.
- Commercial and business aviation product lines delivered strong revenue momentum, expanding 69% year-over-year in the non-defense category during the second quarter of FY2026.
- Nine-month free cash flow expanded 155% year-over-year in FY2026 as the capital expenditure cycle at Exton concluded, transforming operating cash flow conversion.
- Strategic acquisitions, including the closing of the Aydin Displays transaction in July 2026, expanded product scale and diversification while maintaining conservative net leverage of 1.4x.

Our read: Innovative Solutions and Support Inc has completed its operational transition from a capital-intensive manufacturing buildout into a highly profitable, cash-generative avionics leader. With structural gross margins established above 50% and free cash flow accelerating, the shares offer compelling multi-year compounding potential at a substantial discount to intrinsic value.